Vitamins

Definition

vitamins may be defined as potent organic compounds which is found in foods in variable and minute quantity, and must be supplied to the animal organisms from external sources, so that specific physiological functions, vital to life

Characteristics of the Vitamins:

1. The vitamins are widely distributed in nature-both in the animal and vegetable kingdoms
2. Vitamins can perform their work in very low concentration. Hence, the total daily requirement is usually very small.
3. Vitamins can be stored in the body to some extent, for example, the fat-soluble vitamins are stored in the liver and the subcutaneous tissue, vitamin C in adrenal cortex,
4. Vitamins are partly destroyed and are partly excreted.
5. Some vitamins are synthesized in the body, (vitamin A from pro-vitamin carotene and vitamin D from ultra-violet irradiation of vitamin D-precursors, e.g., ergosterol). Some members of the vitamin B complex are synthesized by micro-organisms in the intestinal tract. Vitamin C is also synthesized in some animals e.g., rat.
6. Vitamins are not destroyed in the digestive process and are, therefore, absorbed as such. Hence, all vitamins are effective when administered orally.
7. They do not provide any energy for the animals but they regulate the physiological activities functioning as coenzymes.
8. They processes high specificity of function, each vitamin is related to a definite disease which is caused by its deficiency. Its cure is effected only by supply of the same vitamins

Classification of Vitamins

Vitamins are essential nutrients that your body needs in small amounts to function.

Vitamins are generally classified as water-soluble vitamins and fat-soluble vitamins.

1. Fat-Soluble Vitamins

Vitamin A, D, E and K are fat-soluble. These are stored in adipose tissues and hence are called fat-soluble vitamins.

2. Water-Soluble Vitamins

Vitamins in B-group and vitamin C are water-soluble and cannot be stored in our bodies as they pass with the water in urine. These vitamins must be supplied to our bodies with regular diets.

Characteristics of Fat Soluble Vitamins

Fat-soluble vitamins have specific characteristics that help differentiate them from other vitamins.

Storage: Fat-soluble vitamins are stored in your liver and fatty tissue.

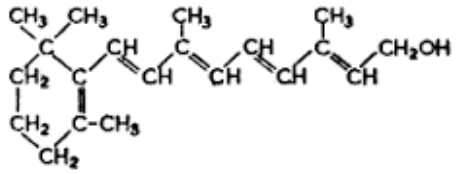
Longevity: Fat-soluble vitamins last longer in your body, because you do not need them every day. They can be tucked away until they are required for such functions as blood clotting, as in the case of vitamin K.

Absorption and Excretion: The fat-soluble vitamins are absorbed through the small intestine with dietary fat and are excreted slowly.

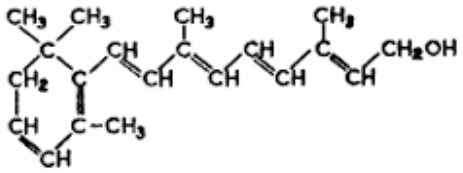
Vitamin A:

Chemical name:

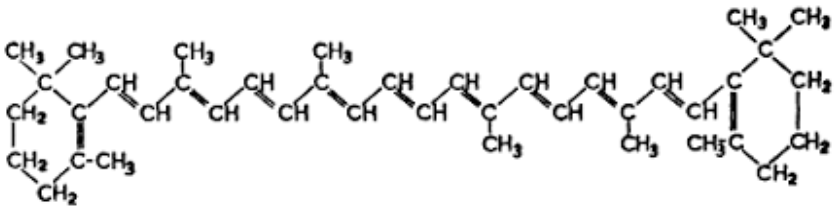
Chemically, vitamin A ($C_{20}H_{29}OH$) is an unsaturated alcohol called retinol. Vitamin A exists in two isomeric forms: vitamin A (retinol) is a trans-isomer and occurs naturally, whereas vitamin A_2 (dehydroretinal or retional2) is a cisisomer and has 30 – 40% of vitamin A actively.



retinol



3-dehydroretinol



β-carotene

Vitamin A_1 aldehyde (retinal) is derived from β-carotene by cleavage at the midpoint of polyene chain connecting the β-ionone rings.

The potency of vitamin A_2 (3-dehydroretinol) is 40% that of vitamin A_1 and it differs from A_1 structurally by the presence of an additional double bond between carbon 3 and 4 of the β-ionone ring.

Sources:

a. Plant Sources:

All pigmented (particularly yellow) vegetables and fruits (e.g., sweet potatoes, carrots, pumpkins, papayas, tomatoes, apricots and peaches) and the leafy green vegetables which supply pro-vitamin A (carotene) in the diet. Cereals also contain carotene.

b. Animal Sources:
Preformed vitamin A (retinol) is supplied by foods of animal origin; they are liver, milk, butter, eggs, kidney, the fat of muscle, meats, and fish liver oil which is very rich in the vitamin.

Functions:

- Retinol (vitamin A aldehyde) combines with lysine residues of opsin protein to form rhodopsin pigments of rod cells of retina, so it is essential for night vision,
- Retinoic acid (vitamin A acid) has some anticancer effects,
- Vitamin A maintains the integrity of epithelial cells; permeability of cell membranes as well as the membranes of organelles,
- In young animals vitamin A causes growth, formation of bones and teeth.

Daily requirements

Adults-5000IU

Pregnant women-6000 to 8000 IU

Children 2000-3000 IU 1 IU=0.3mg retinol 0.6mg β-carotene

Deficiency Symptoms:

Vitamin A deficiency causes the defective night vision called night blindness (nyctalopia or Henerolopia). Vitamin A is essesntial for synthesis of rhodopsin pigment which is very important for vision in dim light. When vitamin A deficient rhodopsin cannt be synthesized .this leads to a failure of vision in dim light

In children deficiency of vitamin A_2 causes xerophthalmia (drying of conjunctiva) keratomalacia (ulceration and softening of cornea) that may lead to complete blindness,
(iii) Toad's skin is another detectable early symptom of vitamin A where the skin becomes dry and rough particularly in the lateral part of forearms and sides of thigh.

Vitamin D:

Clinical name:

Clinically, Vitamin D is called antirachitic vitamin because it possesses the property of curing or preventing ticket.

Chemically Vitamin D is a steroid which is related to calcium metabolism. Hence it is called calciferol. Vitamin D exists in 2 forms i.e. Vitamin D_2 (ergocalciferol) and Vitamin D_3 (cholecalciferol). They are synthesized from the provitamins by the action of UV-rays of sunlight. Because of it, vitamin D is called sunshine vitamin. The provitamin D_2 (ergosterol) occurs in plants and the provitamin D_3 (7-dehydrocholesterol) occurs in animals. The term vitamin D_1 is no longer used.

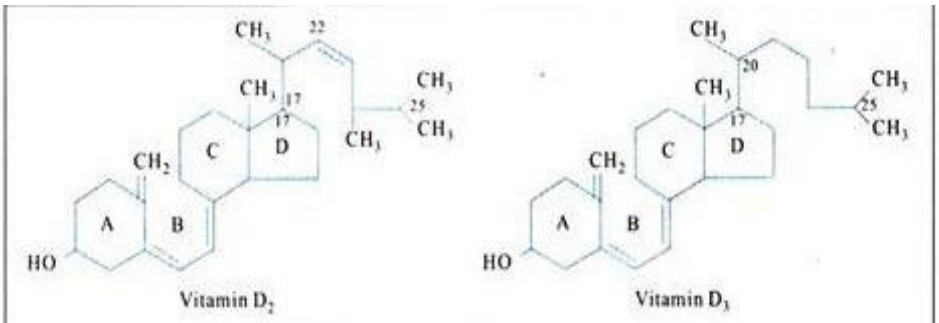


Fig. 15.3 : Structure of vitamin D_2 and vitamin D_3

Ergosterol occurs in plants and 7-de-hydro-cholesterol in animals. Ergosterol is unsaturated and contains an extra methyl group in its side chain than 7-dehydrocholesterol. Ultraviolet irradiation from sunlight cleaves B ring of both compounds. Ergocalciferol (Vit. D_2) is formed in plants and cholecalciferol (Vit D_3) is formed in exposed skin. Both vitamins are of equal potency giving rise to D_2 calcitriol and D_3 calcitriol.
The most important D vitamins are D_2 (activated ergosterol) or ergocalciferol or viosterol and D_3 (activated 7-dehydrocholesterol, cholecalciferol) on the basis of nutrition.
Provitamin D_2 (ergosterol) occurs in the plant kingdom; whereas vitamin D_3 occurs in fish liver oil in nature.

Sources:

Vit. D_3 obtained from fish liver oil, egg, milk, butter, ghee etc.

Daily requirements

Adults-200IU

Pregnant women-400 IU

1IU=0.025 micro grams

Functions:

- Vitamin D_3 is also considered as a prohormone which gives rise to a hormone colcitriol (1, 2, 5 Dihydroxy cholecalciferol) by various metabolic changes. Calcitriol has a role in calcium and phosphate metabolism,
- Vitamin D activates the transcription of mRNA for calcium binding protein,
- It helps in the growth and development of bone and teeth,
- Increase the excretion of phosphate.

Deficiency symptoms:

Vitamin D deficiency leads to rickets in children and osteomalacia in adult.

Osteomalacia

Osteomalacia refers to a marked softening of your bones, most often caused by severe vitamin D deficiency.
When osteomalacia is in its early stages, you might not have symptoms, although signs of osteomalacia might show on an X-ray or other diagnostic tests. As osteomalacia progresses, you might develop bone pain and muscle weakness.
The dull, aching pain associated with osteomalacia most commonly affects the lower back, pelvis, hips, legs and ribs. The pain might be worse at night or when you put pressure on the bones. The pain is rarely relieved completely by rest.

Rickets

Rickets is the softening and weakening of bones in children, usually because of an extreme and prolonged vitamin D deficiency. Rare inherited problems also can cause rickets. Vitamin D helps your child's body absorb calcium and phosphorus from food. Not enough vitamin D makes it difficult to maintain proper calcium and phosphorus levels in bones, which can cause rickets.

Signs and symptoms of rickets can include:

- Delayed growth
- Delayed motor skills
- Pain in the spine, pelvis and legs
- Muscle weakness

Because rickets softens the areas of growing tissue at the ends of a child's bones (growth plates), it can cause skeletal deformities such as:

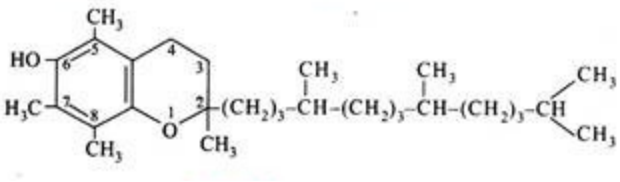
- Bowed legs or knock knees
- Thickened wrists and ankles
- Breastbone projection

Vitamin E:

Clinical name:

It is called antisterility vitamin or fertility vitamin because of its requirement in proper functioning of reproductive system. Tocopherol (tokes = child birth; phero = to bear, ol = alcohol)

Structure



a. Vitamin E refers to a group of compounds known as tocopherols.

b. Four tocopherols namely α-, β-, γ- and δ- have been isolated. β- and γ-tocopherols have two methyl groups in the aromatic nucleus whereas α-tocopherol has three methyl groups in the aromatic nucleus and δ- has only one methyl group.

c. Tocopherols are soluble in fat solvents and destroyed in an alkaline medium. The vitamin activity is destroyed by oxidation.

d. The tocopherols are largely methyl derivatives of the parent compound tocol; they are yellow oily substances.

e. Some tocopherols are derivatives of tocotrienol which contains three double bonds in the aliphatic side chain and is, therefore, terpenoid in structure.

Source:

Vegetable oil, leafy vegetables, milk, cheese, egg, meat etc.

Daily Requirements:

Adults: 25-30 mg.

Function:

- It acts as an antioxidant and prevents oxidation of vitamin A, K, essential fatty acids.

- It keeps the skin glowing by reducing keratinization. Hence, it is also called as beauty vitamin,
- It helps in the normal functioning of skeletal muscles, gonads and renal tubules.
- Vitamin E prevents the per-oxidative changes in membranes of mitochondria and helps in maintaining the smooth translocation of phosphate ions into mitochondria. So the oxidative phosphorylation is enhanced.
- It prevents the hepatic necrosis produced by the lack of sulphur containing amino acids in dietary proteins.
- It prevents development of cerebral disorder.

Deficiency symptomes:

a. Increased susceptibility of erythrocytes to hemolysis by hydrogen peroxide and decrease in the erythrocyte lifespan in human adult males.

b. Anemia, edema and skin changes are observed in infants when fed unsaturated oils.

c. Decreased erythrocyte lifespan, hemolysis, creatinuria and ceriod deposition in malabsorption syndromes in children.

d. Smaller testis and permanent sterility in male rats.

e. Resorption of foetus in female rats.

f. Increased oxygen consumption by skeletal muscle.

g. Hepatic necrosis.

h. Anemia occurs in pregnant and lactating women and in newborn infants due to the deficiency of vitamin E.

i. Its deficiency causes neurologic disorder.

j. Vitamin E deficiency causes muscular dystrophy leading to the increased oxidation of polyunsaturated fatty acids in the muscle with a consequent rise in O_2 consumption and peroxide production which causes an increase in hydrolase activity producing muscular dystrophy and exhibiting increased respiration.
k. Production of nocturnal muscle cramp.

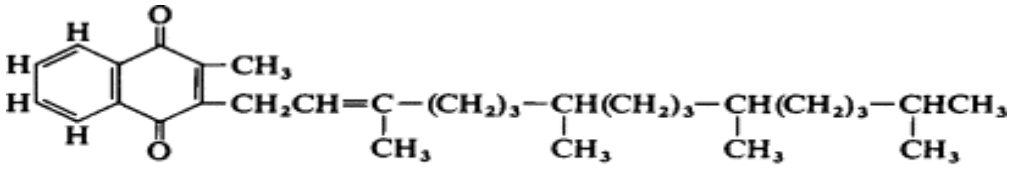
l. Its deficiency causes fibrocystic breast disease and atherosclerosis.

Vitamin K:

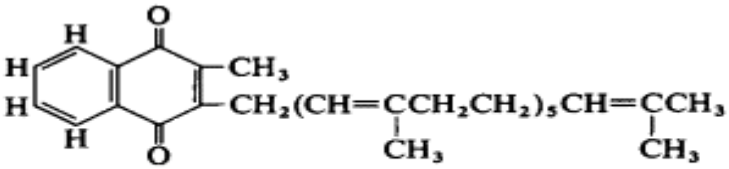
Clinical name:

It is called antihemorrhagic vitamin or vitamin for blood clotting or coagulation vitamin. Chemically vitamin K is a naphthoquinone derivative. Naturally it occurs in two forms, i.e. vitamin K_1 (phylloquinone) and vitamin K_2 (menaquinone or farnoquinone). The vitamin K_3 (mendione) is a synthetic product.

Structure



Vitamin K_1



Vitamin $K_{2(30)}$

a.The several substances (natural and synthetic) with vitamin K (anti-hemorrhagic) activity are naphthoquinones.

b. Two naturally occurring vitamins K have been identified. Vitamin K was isolated originally from alfalfa and vitamin K_2 was originally isolated from putrid fish meal. Menadione is the most important of the synthetic vitamins K.
c. The natural vitamins K are “fat-soluble” vitamins. The synthetic vitamins K are lacking the long hydrocarbon chain and hence are soluble in water to some extent.

Source:

Green leafy vegetables, soybean, carrots, potatoes, milk, fish and meats etc.

Daily Requirements:

The average diet contains adequate amounts of vitamin K_1 and K_2 being synthesized by bacteria in the intestine. So the vitamin K deficiency has not been reported in healthy individuals except in newborn infants fed on mother's milk when mother's diet has low vitamin K content.

Physiological Functions:

a. Vitamin K catalyzes the synthesis of prothrombin by the liver after transcription from information carried on messenger RNA.

b. It reduces the prothrombin time.

c. It. regulates the synthesis of plasma clotting factors (factors VII, IX and X).

d. Vitamin K_1 is an essential component of the phosphorylation processes involved in photosynthesis in green plants. It is also involved in oxidative phosphorylation in animal tissues.
e. Vitamin K is involved in the maintenance of normal levels of blood clotting factors II, VII, IX and X, all of which are synthesized in the liver initially as inactive precursor proteins.

f. Vitamin K acts as cofactor of the carboxylase that forms γ-carboxyglutamate residues in precursor proteins.

g. Vitamin K is used therapeutically as an antidote to poisoning by dicoumarol-type drugs. The quinone forms of vitamin K will bypass the inhibited epoxide reductase and provide a potential source of the active hydroquinone form of vitamin K.

h. Vitamin K acts as a cofactor of carboxylase.

i. Vitamin K is required for absorption of fat.

Deficiency symptomes:

a. The deficiency of Vit. K leads to a lowering of prothrombin level and increased clotting time of blood. This may lead to hemorrhagic conditions.

b. Vitamin K deficiency causes hemorrhagic disease of the newborn.

c. Vitamin K deficiency is caused by fat malabsorption which may be associated with pancreatic dysfunction, biliary disease, atrophy of the intestinal mucosa or any cause of steatorrhea.

d. Sterilization of the large intestine by antibiotics can result in deficiency when dietary intake is limited.

e. Short circuiting of the bowel as a result of surgery may also cause deficiency which may not respond even to large doses of vitamin K. Water-soluble form of vitamin K (vitamin K_3) is alone useful in such cases.

WATER SOLUBLE VITAMINS

A vitamin that can dissolve in water. Water-soluble vitamins are carried to the body's tissues but are not stored in the body. They are found in plant and animal foods or dietary supplements and must be taken in daily. Vitamin C and members of the vitamin B complex are water-soluble.

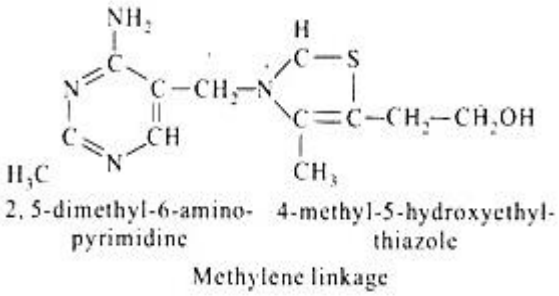
Vitamin B-Complex:

It includes at least 10-11 different water soluble vitamins. They are grouped together as B-complex because all the members' acts as coenzymes and their actions are closely related.

Vitamin B1- Thamine

Vitamin B1, also called thiamine, is a B complex vitamin On account of its curing action against beri beri. It is commonly known as antiberiberi factor . According to the University of Maryland Medical Center (UMM), thiamine was named B1 because it was the first B complex vitamin to be discovered.

Structure



Thiamine is readily soluble in water. It is stable in acid medium. It is destroyed when autoclaved at 120°C for 30 minutes. It is destroyed even at room temperature in an alkaline medium**.**

Sources

There are many natural ways to add thiamine-rich foods to an everyday diet. Food sources of thiamine include beef, liver, dried milk, nuts, oats, oranges, pork, eggs, seeds, legumes, peas and yeast. Foods are also fortified with thiamine. Some foods that are often fortified with B1 are rice, pasta, breads, cereals and flour.

Dosage

- Males who are 14 and older should consume 1.2 milligrams per day (mg/day).
- Females who are 14 to 18 years old should consume 1.0 mg/day.
- Females who are 19 and older should consume 1.1 mg/day, though women who are pregnant or breastfeeding may need more and should consult her doctor.

Functions

- Thiamine is essential for growth.
- It is essential for maintaining the nerves in normal condition.
- It undergoes reaction with ATP in which two terminal phosphate from ATP are transferred to form thiamine pyrophosphate(TPP). TPP acts as a coenzyme in glycolysis pathway as well as in krebs cycle.

Deficiency Manifestation:

I. Beriberi:
It is a nutritional disorder i.e. thiamine deficiency.

Three forms of disease are:
(a) Wet beriberi:
(Oedema is the most prominent feature and develops rapidly in the legs, face etc.
Anorexia and dyspepsia are present.
The calf muscles are tense, slightly swollen and tender on pressure (pins and needles).
The pulse is fast.
The apex beat of heart is displaced outwards.

(b) Dry beriberi:
The essential feature is polyneuropathy.
The subject needs stick to stand and walk and finally becomes bedridden.
The concentrations of pyruvic acids and lactic acids are increased.
Transketolase activity of red cells is decreased.
The muscles become wasted and weak and it is difficult to walk.

(c) Infantile beriberi:
This occurs in breast-fed infants between 2nd to 5th months.
In acute form—Cardiac failure develops, the mother notices that infant is restless, cries a lot, is passing less urine, than normal. The infant may be cyanosed with dyspnoea and tachycardia and may die within 24 to 48 hours.
In chronic form—the main symptoms are due to gastrointestinal disturbances. There is constipation and vomiting. The child is fretful and sleeps poorly.
Polyneuritis in birds

Polyneuritis may be seen in mature birds ~3 wk after they are fed a thiamine-deficient diet. As the deficiency progresses, birds may sit on flexed legs and draw back their heads in a star-gazing position. Retraction of the head is due to paralysis of the anterior neck muscles. Soon after this stage, chickens lose the ability to stand or sit upright and topple to the floor,

VitaminB2(Riboflavin)

It was first isolated from milk hence the erlier name was Lactoflavin. It was also known as ovaflavin(from egg) ans hepatoflavin(from liver). It is popularly called as the yellow enzyme because of its colour

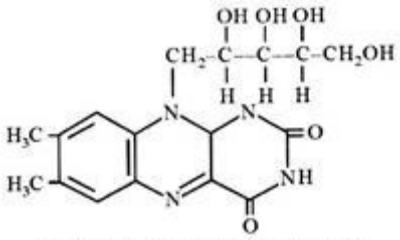
Structure

It contains an isoalloxazine nucleus.

It is water-soluble, heat-stable and sensitive to light.

d. The solution of vitamin B_2, when exposed to ultraviolet light, emits a strong greenish yellow fluorescence.
e. On oxidation with ultraviolet rays or visible light, it undergoes irreversible decomposition.

f. It is rapidly destroyed when the solution is exposed to bright light.



6, 7-dimethyl-9-D-ribitolisoalloxazine

Source

Riboflavin is widely distributed throughout the plant and animal kingdom

Rich source=anaerobic fermentating bacteria

Good source-Milk Liver Kidney heart fish and leafy vegetables are excellent

Fair source-cereals roots vegetables germinating wheat and barley

Daily requirements

Infants 0.4-0.6mg

Children-0.8-1.2mg

Adult male 1.5-1.8

Adult female 1.1-1.4mg

Pregnant and lactating wokmen1.5-1.9mg

Functions:

1 Riboflavin is involved in the regulatory functions of some hormones connected with carbohydrate metabolism.

2 The free riboflavin present in retina is converted by light to a compound involved in stimulation of the optic nerve.

3.Riboflavin is a component of two importantant co enzymes namely
a. FMN — Flavin mononucleotide
b. FAD — Flavin adenine dinucleotide

4. It acts as a coenzyme for enzyme catalysed oxidation-reduction reaction.

5. It is the component of aerobic dehydrogenase.

6.This is one of the component of respiratory chain for the trans-part of electron:

Deficiency disease:

a. Lips—Redness and shiny appearance of lips.

b. Cheilosis—Lesions at the mucocutaneous junction at the angles of mouth leading to painful fissures.

c. Tongue—Painful glossitis, the tongue assuming a red purple (magenta) colour.

d. Seborrheic dermatitis—Scaly, greasy, desquamation chiefly about the ears, nose and nasolabial folds.

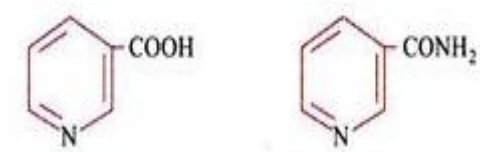
e. Eyes—May lead to corneal vascularization and inflammation with cloudiness of cornea, watering, burning of eyes, photophobia and cataract.

Vit. B_3-Niacin (Nicotinic Acid):

During 1922-1928 Goldberger and coworkers treated the disease pellagra in human beings Goldberger named niacin as the Pellagra-Preventive (P-P) factor. As this vitamin ha a curing avtion against Black tongue disease in Dogs it is also called as antiblack tongue factor

Structure

Niacin is pyridine 3-carboxylic acid. It occurs in tissues as niacin amide (nicotinamide) It is soluble in water. It is stable to heat and not destroyed by autoclaving at 120°C for 20 minutes in acid or alkaline medium Excess nicotinic acid is not stored in the body.



Nicotinic acid Nicotinamide (Niacinamide)

Source

Richest source: yeast, rice polishing

Good source: Meat liver and poultry

Fair source: milk , eggs, tomatoes, leafy green vegetable

Poor source: Most fruits and vegetables

Functions:

- Nicotinic acid is essential for the normal functioning of the skin, intestinal tract and the nervous system.

- It is essential for growth
- It prevents pellagra(rough skin)
- Nicotinic acid is used therapeutically for lowering plasma cholesterol.
- It is a component of two coenzymes NAD (di-hydro-nicotinamide adenine) and NADP(di-hydro-nicotinamide adenine dinucleotide phosphate). These coenzymes play an important role in metabolism by acting as hydrogen and electron transfer agents by means of reversible oxidation and reduction.
- Neacin has a great importance in human nutrition as well as in the requirements of many other organisms including bacteria and yeast.

Deficiency Disease:

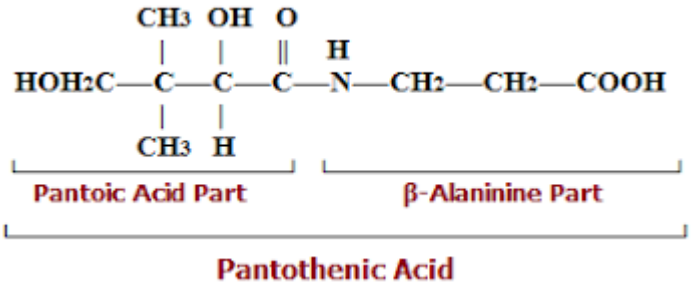
The deficiency of niacin causes the disease pellagra. The clinical features of the disease include three D'S-dermatitis (lesion of skin of face, neck, knees, breasts, thick and scaly skin), diarrhoea, dementia (headache, depression, anxiety, insomnia and forgetfulness).

Vitamin B5-(Panthonic acid)

Williams and co-workers isolated this vitamin in a pure form as its calcium salt.
On account of its wide distribution it is called as panthonic acid (panthos means everyehere)

Structure:

Pantothenic acid consists of β-alanine and pantoic acid joined through a peptide bond. It is highly soluble in water. It is stable to autoclaving at 120°C for 30 minutes in neutral solution but is destroyed in acid or alkaline medium.



Source

Richest source: Jelly

Good source: yeast liver rice polishing and wheat germ

Fair source: Meat ,milk , eggs, and leafy vegetables

Poor source: fruits and vegetables

Daily requirements

Infants: 1-2mg

Children:4-5mg

Adults:5-10mg

Pregnant and lactating women-10-15mg

Functions

- Pantothenic acid is essential for the growth of infants and children
- Pantothenic acid as a constituent of coenzyme A is required for several fundamental reactions in metabolism.
- significant amount of the cellular pantothenic acid is protein-bound. This form is contained in a compound known as acyl carrier protein, a coenzyme required in the biosynthesis of fatty acids.

Deficiency Symptoms:

Deficiency of this vitamin in humans results in nausea, vomiting, certain gastrointestinal disorders, irritability, inadequate growth, anemia, fatty liver, failure in gaining weights.

Pantothenate deficiency causes burning foot syndrome in prisoners of war and is associated with reduced capacity for acetylation.

Vitamin B6(Pyridoxine)

It is also named as adermin or antidermatitis factor. It includes 3 compounds Pyridoxine, pyridoxal and Pyridoxamine

Structure:

The more active derivatives are pyridoxal and pyridoxamine phosphates. Pyridoxine is readily soluble in water. It is readily absorbed from the small intestine. The excess amount, if ingested, is not stored in the body but is excreted in urine.

Daily requirements

Infants: 0.3mg

Children:0.6-1.2mg

Adults:1.6-2,0mg

Pregnant and lactating women-2.5mg

Functions:

a. Pyridoxine is essential for the growth of infants.

b. pyridoxial phosphate acts as a coenzyme

It is involved in the active transport of amino acid and certain metabolic ions across cell membrane

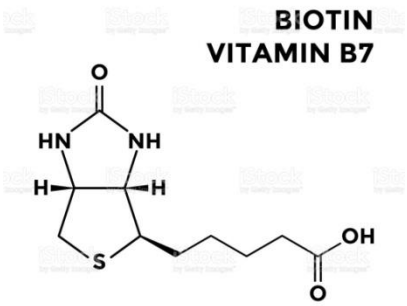
Deficiency Symptoms:

Fortunately, deficiency of vitamin B_6 is rare because of its easy availability in most foodstuffs and intestinal bacteria are capable of synthesizing vit-B6
Deficiency gives rise to irritability and depression.
Deficiency of this vitamin occurs in infants on inadequate milk formulas.
The major symptoms are convulsions due to depletion of brain

Vitamin B7 (Biotin)

It is also called as anti egg white injury factor. Vitamin B7 also known as bioti is one of the water soluble and highly essential B complex vitamins. Termed as Vitamin H where the letter H represents Haar and Haut, which are german words for “hair and Skin”

Structure



Source

Biotin is widely distributed in natural foods. Intestinal bacteria also supply biotin in large proportions

Richest source: Egg, yolk , kidney molasses

Good source: yeast, milk, tomatoes

Fair source: fruits and vegetables and human milk

Functions

Biotin is essential for the body's metabolism. It acts as a coenzyme in a number of metabolic pathways involving fatty acids and essential amino acids, as well as in gluconeogenesis

It helps in conversion of pyruvic acid to oxalo acetic acid

Succinic dehydrogeninase succinic acid decarboxylase aspartic acid deaminase serine deaminase etc are influenced by Biotin

Daily requirements

Infants: 10-15 µg

Children:20-40 µg

Adults:50-60µg

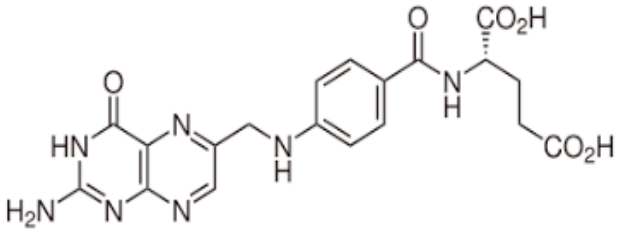
Deficiency symptomes

Although quite rare, biotin deficiency mostly occurs due to a lack of this quintessential ingredient from the daily diet. Regular consumption of egg white for several months can lead to biotin deficiency due to the avidin content in eggs. Avidin is a type of protein found in eggs, which binds tightly with the biotin in the intestines and prevents them to get absorbed by the body or function properly.
Some of the signs of biotin deficiency are characterized by loss of hair, dry brittle nails, scaly, red rash around the eyes, nose, and mouth, dry eyes, scaly dry skin causing dermatitis, conjunctivitis, sore tongue, nausea, vomiting and loss of appetite. Other symptoms are related to psychological or neurological problems, which include depression, types of insomnia, hallucinations, lethargy, numbness and tingling sensation in the arms and legs.

Vitamin B9 (Folic Acid)

This vitamin was obtained from spinach leaf and this led to its nomenclature as folic acid. The official name of this vitamin is folacin

Structure



Folic acid (folacin, pteroylglutamic acid) is a compound made up of the Pteridine nucleus, P-amino-benzoic acid and glutamic acid. Folic acid is soluble in water

Source

Richest source: yeast, Liver Kidney, wheat

Good source: Meat, fish green leaf vegetables

Fair source: milk, potato rice tomato banana etc

Functions

It functions as a coenzyme

Its main role is in the formation and maturation of red blood cells

It takes part in the synthesis of purines, hence they are fundamentally involved in the growth and reproduction of cells

Daily requirements

Infants: 50 µg

Children:100-300µg

Adults:400µg

Pregnant women-800 µg

Deficiency

The deficiency gives rise to a megaloblastic anemia. The nuclei of the neutrophil polymorpho-nuclear leukocytes contain more than the normal number of lobes. Other deficiency manifestations include retardation of growth, weakness, infertility, inadequate lactation in females

Vitamin B_{12} (Cyanocobalamin)

Vitamin B_{12} is a member of the vitamin B complex. Vitamin B_{12} contains cobalt and therefore is also known as cobalamin.

Vitamin B_{12} was discovered from its relationship to the disease pernicious anemia, which is an autoimmune disease that destroys parietal cells in the stomach that secrete intrinsic factor

Source

Vitamin B12 is present only in the foods of animal origin. It is not present in plants, Bacterial synthesis of cobalmine occurs in human colon but it is not obsorbed
Richest source: Liver Kidney,

Good source: Meat, fish

Fair source: milk, cheese

Functions

- It is essential for the development of RBCs
- It is involved in the synthesis of nuclic acids
- It stimulate the appetite and general health
- It cures the symptoms ofperinecious anemia
- Coenzyme B12 is involved in the biosynthesis of proteins
- Co enzyme B12 is involved in the synthesis of methyl group

Daily requirements

Infants: 0.3µg

Children:1-2µg

Adults:3µg

Pregnant women-4µg

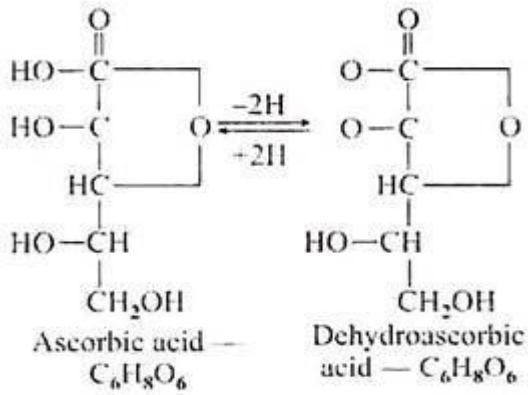
Deficiency

It is exclusively synthesized by bacteria and is found primarily in meat, eggs and dairy products. Vitamin B_{12} is necessary for the synthesis of red blood cells, the maintenance of the nervous system, and growth and development in children.
Deficiency of vitamin B_{12} can cause pernicious anaemia,(charactersed by the drastic decrease in the blood cell count). RBC becomes abnormally large, haemoglobin content gets low.
vitamin B_{12} neuropathy, (degeneration of nerve fibres and irreversible neurological damage).

Vitamin C(Ascorbic acid)

It is one of the important biochemical redox system. Oxidation of ascorbic acid gives rise to dehydroascorbic acid. This reaction is reversible. Both forms are physiologically active. This reaction has provided the basis of oxidation and reduction process (biological oxidation).

Structure



It is white crystal, soluble in water, (insoluble ion fat-sol-vents) and heat-labile. Easily oxidizes at 100° C in presence of oxygen.

Source

Fresh fruits, mainly citrus fruits, e.g., emblic myrobalan (amlaki), orange, lemon, tomato, pine-apple, papaya, etc., fresh vegetables, e.g. cabbage, cauliflower, lettuce, spinach, green peppers, beans etc.; properly sprouted pulses, germinating grams etc., are also rich sources. Potato and seeds are poor, but rich during germination. In the animal body the adrenal cortex contains a good amount. Cow's milk, meat and fish contain a little.

Functions:

a. Since it is very sensitive to reversible oxidation (ascorbic acid ⇋ dehydroascorbic acid) it is involved in the oxidation-reduction reactions of the cell.

b. It is involved in the conversion of folic acid to folinic acid (citrovorum factor).

c. It is involved in the hydroxylation of steroids in the adrenal cortex.

d. It is required in the metabolism of tyrosine and phenylalanine and also in tryptophan.

e. It is required for the absorption of iron and incorporation of plasma iron in ferritin.

f. It is involved in the formation of norepinephrine.

g. It is essential for the normal regulation of the colloidal condition of intercellular substances including the fibrils and collagen of connective tissue, osteoid tissue, dentine, the intercellular ‘cement substance’ of the capillaries.

h. It is concerned in the hydroxylation of proline and hydroxyproline which is an important constituent of collagen.

i. It has an inhibitory effect on the hyaluronidase-hyaluronic acid system.

j. High doses of vitamin C prevent the common cold or reduce the duration of its symptoms.

Deficiency Diseases:

Severe ascorbic acid deficiency produces scurvy. The signs of this deficiency are entirely confined to supporting tissues of mesenchymal origin (bone, dentine, cartilage, and connective tissue). Scurvy is characterized by failure in the formation and maintenance of intercellular materials which causes typical symptoms such as:

a. Internal haemorrhages.

b. Loosening of the teeth.

c. Poor healing of wounds.

d. Swelling of long bones.

e. Easy fracturability of bones.

f. Swelling, sponginess, tenderness and bleeding of gums.

g. Anemia.

h. Susceptibility to infections.

i. General weakness.